Bo Li | 李波

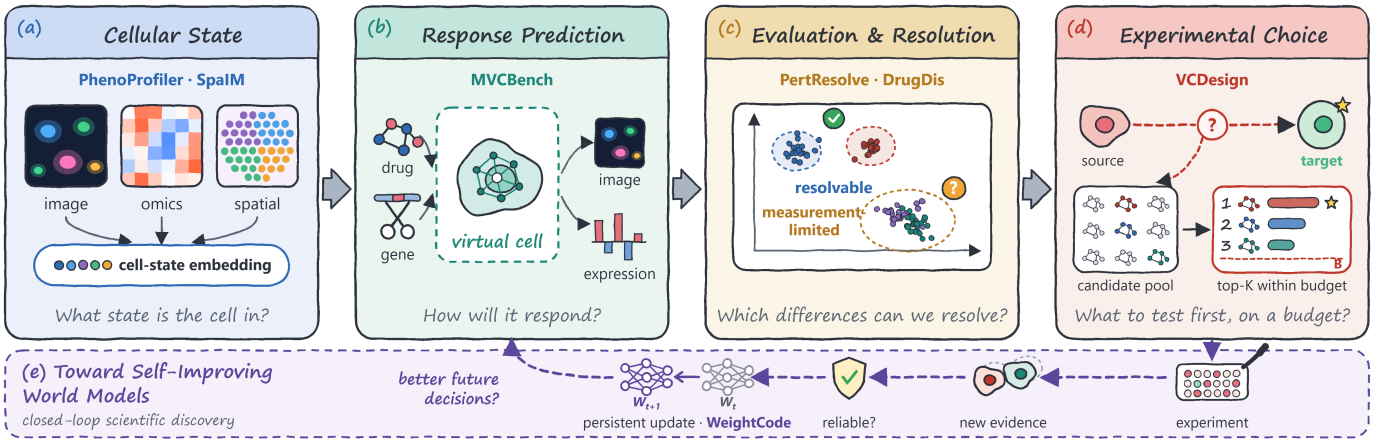
Ph.D. Candidate in Computer Science, University of Macau
Visiting Student, National University of Singapore

Virtual-Cell World Models · **Closed-Loop Design** · **Self-Improving Models**

Email
Homepage
Google Scholar
GitHub
ORCID

RESEARCH PROFILE

I study how AI systems can learn from observations, guide scientific interventions, and improve through new evidence. My research spans multimodal cellular modeling, genetic and drug perturbation prediction, measurement-aware evaluation, intervention design, and methods for expanding the capabilities of pretrained models. My long-term goal is to develop scientific AI systems that turn new knowledge and experimental feedback into lasting model capabilities and better intervention decisions.



EDUCATION & APPOINTMENTS

University of Macau — Aug 2024 – Jun 2027 (expected)
Ph.D. in Computer Science | Advisor: Prof. Bob Zhang; Co-advisor: Prof. Qianqian Song

National University of Singapore — Jun 2026 – Jun 2027
Visiting Ph.D. Student | Host: Prof. Yang Zhang

Beijing University of Technology — Sep 2021 – Jul 2024
M.Eng. in Electronic Information | Advisors: Prof. Yong Zhang and Prof. Baocai Yin

Beijing Information Science & Technology University — Sep 2017 – Jul 2021
B.Eng. in Robotics Engineering | Advisor: Prof. Hongbo Huang

10	3	590	1,200+
First-author journal articles	First-author *Nature Communications*	citations · h-index 15 Google Scholar · Nov 2026	**GitHub stars** Across open-source projects

RESEARCH PROGRAM & SELECTED PUBLICATIONS

9 core works · † equal contribution

CELLULAR STATE PhenoProfiler · SpaIM

What representations of cellular state can be learned from imaging and omics measurements?

1. **PhenoProfiler: Advancing phenotypic learning for image-based drug discovery**
   Bo Li, Bob Zhang, …, Mengran Li, Yong Zhang, Qianqian Song · *Nature Communications* 17, 793 (2026)
   Learns phenotypic profiles end to end from whole high-content microscopy images, for compound profiling.
2. **SpaIM: Single-cell spatial transcriptomics imputation via style transfer**
   Bo Li†, Ziyang Tang†, …, Baijian Yang, Jing Su, Qianqian Song · *Nature Communications* 16, 7861 (2025)
   Treats imputation of unmeasured genes in spatial transcriptomics as style transfer from single-cell references.

RESPONSE PREDICTION MVCBench

Do learned cell-state transitions under intervention generalize across modalities, compounds and contexts?

3. **MVCBench: A multimodal benchmark for drug-induced virtual cell phenotypes**
   Bo Li†, Qing Wang†, Shihang Wang†, …, Xiaojun Yao, Chuxia Deng, Qianqian Song · *Nature Communications* (in press)
   Benchmarks image- and expression-based perturbation prediction under compound, cell-line, plate and dataset shifts.

EVALUATION & RESOLUTION PertResolve · DrugDis

When does a predicted gain reflect biology that the assay can reproducibly resolve?

4. **Measurement resolution constrains fine-grained perturbation prediction**
   Bo Li†, Chengyang Zhang†, Mengran Li†, …, Yuhao Yi, Jiancheng Lv, Yang Zhang · *Under review* (2026)
   Tests which perturbation distinctions an assay can resolve, separating model limits from measurement limits.
5. **Disentangling general and context-specific effects in drug-response prediction**
   Bo Li†, Chengliang Liu†, Yuzhong Peng†, …, Shenghui Huang, Chuxia Deng, Yang Zhang · *Under review* (2026)
   Separates general drug, general sample and drug–sample interaction effects that aggregate accuracy can hide.

EXPERIMENTAL CHOICE VCDesign

Planning with the model: given a target cellular state and a finite budget, which interventions are worth testing next?

6. **VCDesign: Finite-budget intervention design for virtual cells**
   Bo Li†, Lin Wang†, Bob Zhang, …, Chengliang Liu, Zhiyuan Liu, Yang Zhang · *Under review* (2026)
   Ranks candidate interventions toward a target state under a fixed budget, scores the selected set against independently measured held-out outcomes, and transfers evidence to candidates with no measured response.

SELF-IMPROVING MODELS WeightCode · WeightLoop · CellScientist

Closing the loop: can measured outcomes keep adding capabilities to a model, and do the gains improve its next decisions?

7. **WeightCode: Installing new capabilities into neural networks via weight-space programming**
   Bo Li†, Taoyong Cui†, Bob Zhang, …, Zhenfei Yin, Yingcheng Wu, Ling Yang · *Under review* (2026)
   Weight programs install capabilities without gradients (SCAN 0→99.8%; BabyAI pickup 0→98.4% from 8 demos); rerun on five independently trained ViTs they keep 83–110% of their gains; new programs extend edited networks.
8. **WeightLoop: Weight-space programming with experimental feedback**
   Mengran Li, **Bo Li**, Weijing Zhao, …, Victor C. M. Leung, Stan Z. Li · *arXiv* 2611.07319 (2026)
   A reusable editing program turns new assay results into persistent updates of a deployed predictor, without retraining.
9. **CellScientist: From execution feedback to auditable model-revision trajectories for cellular perturbation prediction**
   Mengran Li†, **Bo Li**†, Jiaying Wang, …, Stan Z. Li, Zelin Zang · *arXiv* 2605.07335 (2026)
   Turns execution feedback into auditable model revisions, finding better held-out predictors at small budgets.

NEXT DIRECTIONS

- **Updatable cell world models.** Turn new assay results into persistent, transferable model updates, not retraining.
- **Does capability growth pay off?** Test whether each update improves intervention choices on held-out experiments.
- **Accumulating capabilities.** Reusable model edits that add perturbation and cell-context skills without forgetting.

OTHER PUBLICATIONS

full list of 37 works on Google Scholar

Co-first- and co-corresponding-author manuscripts

10. **When do single-cell response distributions improve drug retrieval?**
    Huihong Wang, Xiaoping Liu, Chen Chen, Qiuyan Sun#, **Bo Li#**, Bob Zhang# · *Under review* (2026)
11. **When label-free morphology is sufficient for targeted cellular measurements**
    Mengran Li†, Jianqing Zhu†, **Bo Li**†, …, Lian Zhang, Jinchao Xu · *bioRxiv* (2026)

Earlier first-author journal papers: *Pattern Recognition 2024* · *EAAI 2024 (×2)* · *ACM TOMM 2023* · *Visual Computer 2023*

OPEN SOURCE & SKILLS

1,200+ GitHub stars in total

Nature-Paper-Skills	Agent skills for scientific writing, revision, auditing and resubmission.	**670 stars**
Awesome Virtual Cell	Research landscape: 552 papers · 20 themes · interactive map.	**336 stars**
DrugClaw	Agentic RAG for drug research · 57 skills · 15 task categories.	**118 stars**

Skills: PyTorch · LLM agents · weight-space editing · multimodal learning · high-content imaging · single-cell and spatial omics

TALKS, MENTORING & SERVICE

Invited talk: PhenoProfiler, Prof. Guangyu Wang's group, Weill Cornell Medicine & Houston Methodist — Feb 2025

Mentoring: co-mentored four BJUT graduate students; six first-author papers, incl. New Phytologist, IEEE JBHI, ACM TOMM.

Reviewer: Science Advances; IEEE TPAMI, TIP, TNNLS; Medical Image Analysis; Bioinformatics; Briefings in Bioinformatics; Pattern Recognition; BMC Biology; Eng. Appl. Artif. Intell.; Expert Syst. Appl.; ICLR 2027

HONORS, AWARDS & PATENTS

National Scholarship, China's top graduate award	2022	**Beijing Outstanding Graduate**	2021, 2024
Top 100 Graduates, BJUT	2024	**Xiaomi Scholarship**	2023
First-Class Academic Scholarship, BJUT	2022, 2023	**Ph.D. Full Scholarship**, UM	2024

Patents: two granted Chinese invention patents on cell localization, co-inventor (2026): CN115457547B · CN115810046B